

Project Context Block

Efficient Context & Agent Tool Generator

1. Project Goal

Build a local program that converts long Korean project notes, development explanations, and rough context into token-efficient AI-agent-ready files.

The program should help generate and manage:

- AGENTS.md
- TASK.md
- CONTEXT.raw.ko.md
- CONTEXT.compact.md
- CONTEXT_FOR_AI.md
- .agents/skills/*/SKILL.md
- optional JSON context files
- optional project tree summaries
- optional data/index summaries

The main target AI tools are:

- Codex
- Antigravity
- other AI coding agents that benefit from structured project context

The program should not change the user's development structure. It should support the existing workflow only.

2. Fixed Development Workflow

The user's development structure is fixed and must not be changed.

Source / active development:

- Google Drive mirrored workspace

Backup / deployment:

- GitHub repository

Runtime:

- Local machines pull updates from GitHub using `git pull`

Important rule:

Do not suggest, create, or require a separate clone or worktree.
The Google Drive mirrored workspace is the active development source of truth.
GitHub is for backup, version tracking, deployment, and distribution.
Runtime folders are consumers of GitHub updates.

3. Core Concept

The program should act as a “Context Compiler.”

Input:

Long Korean notes
rough project explanations
development workflow descriptions
raw context
task requests
folder structure
project-specific rules

Output:

short, structured, token-efficient English or mixed-language Markdown
agent-facing rule files
task files
skill templates
context bundles

Basic pipeline:

```
CONTEXT.raw.ko.md
↓
context classification
↓
deduplication
↓
translation / compression
↓
structured JSON
↓
Markdown rendering
```

↓

AGENTS.md / TASK.md / CONTEXT.compact.md / CONTEXT_FOR_AI.md

4. Preferred Language System

For AI coding agents, the most efficient format is:

```
Controlled Technical English
+ Markdown headings
+ short bullet points
+ explicit constraints
+ explicit forbidden actions
+ explicit verification commands
+ explicit output format
```

Korean can be used for raw notes and domain explanations, but agent-facing files should prefer concise technical English.

Recommended structure:

```
Goal
Context
Scope
Constraints
Forbidden
Procedure
Verification
Output Format
```

5. Main Files and Their Roles

```
AGENTS.md
= project-level permanent rules

TASK.md
= current task instruction

CONTEXT.raw.ko.md
= long Korean raw notes

CONTEXT.compact.md
= compressed AI-facing context
```

```
CONTEXT.compact.json
= structured machine-readable context

CONTEXT_FOR_AI.md
= final context bundle for AI agents

.agents/skills/*/SKILL.md
= reusable workflow manuals

scripts/build_context.py
= builds final context bundle

scripts/context_compiler_llm.py
= local LLM-based context converter

scripts/context_rule_based.py
= Python-only rule-based context converter
```

6. AGENTS.md Requirements

The program should be able to generate an `AGENTS.md` file with the following sections:

```
Project Workflow
Editing Rules
Google Drive Safety
Data Safety
Scientific Analysis Rules
Dependency Policy
Git Rules
Verification
Final Report Format
```

The generated `AGENTS.md` must include these fixed workflow rules:

Project Workflow

- The source of truth is the Google Drive mirrored workspace.
- Active development happens directly in the Google Drive mirrored workspace.
- GitHub is used for backup, version tracking, deployment, and distribution.
- Runtime machines receive code updates through ``git pull`` from GitHub.
- Do not assume any runtime folder is the primary development folder.
- Do not propose changing this workflow to a separate clone or worktree.

Recommended editing rules:

Editing Rules

- Make minimal, focused changes.
- Inspect relevant files before editing.
- Do not refactor unrelated code.
- Do not rename files, folders, public functions, config keys, or output columns unless explicitly requested.
- Preserve the current project structure.
- Preserve existing Korean comments and UI text unless the task is specifically about language cleanup.
- Use English for technical identifiers and agent-facing documentation.

Recommended Google Drive safety rules:

Google Drive Safety

- Assume files are synchronized by Google Drive.
- Avoid rapid mass file creation, deletion, or renaming.
- Do not create unnecessary duplicate files.
- Do not overwrite large files blindly.
- Do not modify Google Drive sync settings.
- Do not move the project folder.
- If a file conflict or duplicate sync file is detected, report it before editing.

Recommended data safety rules:

Data Safety

- Treat raw experiment data as read-only.
- Never delete or overwrite files under:
 - `data/`
 - `raw/`
 - `backup/`
 - `archive/`
 - `results/raw/`
- Never modify original `.txt`, `.bmp`, `.jpg`, `.png`, `.csv`, or measurement files unless explicitly requested.
- Derived outputs must be written only to configured output folders.
- Preserve original filenames in output metadata when possible.

Recommended verification rules:

Verification

After changes, run the smallest relevant verification command.

Prefer these checks when applicable:

- Python syntax check: ``python -m compileall.``
- Test command: ``pytest``
- GUI startup command: ``python main.py``
- Script dry run if available

If verification cannot be run, explain why.

Recommended final report format:

Final Report Format

Always report:

- root cause
- changed files
- commands run
- verification result
- remaining risks

7. TASK.md Requirements

The program should generate a reusable `TASK.md` template.

Recommended structure:

TASK.md

Fixed Workflow

Use the existing workflow only:

- Source/development: Google Drive mirrored workspace
- Backup/deployment: GitHub repo
- Runtime: local machines pull from GitHub

Do not suggest or create a separate clone, worktree, or alternative development structure.

Goal

Describe the specific task here.

Scope

- List files, modules, or behavior that may be changed.
- Keep the task narrow.
- Do not refactor unrelated code.

Current Problem

Describe the observed issue or requested feature.

Constraints

- Preserve existing workflow.
- Preserve existing file structure.
- Do not modify raw data.
- Do not add dependencies unless necessary.

Verification

- Run the relevant command.
- Check expected behavior.
- Report failure clearly if verification cannot be completed.

Output

- Explain root cause.
- List changed files.
- List commands run.
- Report verification result.

8. CONTEXT.compact.md Requirements

The program should convert rough context into concise sections.

Recommended sections:

Project
Environment
Workflow
Data Rules
Coding Rules

Forbidden
Verification
Output Format

Example output:

Compact Context

Workflow

- Source of truth: Google Drive mirrored workspace.
- GitHub is used for backup, version tracking, deployment, and distribution.
- Runtime machines receive updates through ``git pull``.
- Do not treat runtime folders as primary development folders.
- Do not suggest clone or worktree alternatives.

Data Rules

- Raw experiment data is read-only.
- Do not delete or overwrite raw data.
- Derived outputs must go to configured output folders.

Coding Rules

- Make minimal focused changes.
- Avoid unrelated refactoring.
- Preserve existing file paths and output naming conventions.

Verification

- Run the smallest relevant test or startup command.
- Report changed files and commands run.

9. CONTEXT_FOR_AI.md Requirements

The program should generate a final bundled file for AI agents.

Recommended bundle order:

1. AGENTS.md
2. CONTEXT.compact.md
3. TASK.md
4. Project Tree

- 5. Optional important file summaries
- 6. Optional generated indexes

The final AI prompt should be short:

```
Read CONTEXT_FOR_AI.md.  
Follow AGENTS.md strictly.  
Complete TASK.md only.  
Use the existing Google Drive → GitHub → git pull workflow.  
Do not suggest clone or worktree alternatives.
```

10. Skill System Requirements

The program should generate reusable skill folders.

Recommended structure:

```
.agents/  
└─ skills/  
    │   └─ code-review/  
    │       └─ SKILL.md  
    │   └─ safe-refactor/  
    │       └─ SKILL.md  
    │   └─ git-backup/  
    │       └─ SKILL.md  
    │   └─ context-compiler/  
    │       └─ SKILL.md  
    └─ make-analysis/  
        └─ SKILL.md
```

Skill meaning:

```
Skill  
= reusable workflow manual for a specific task  
  
AGENTS.md  
= permanent project rules  
  
TASK.md  
= one-time current task instruction  
  
MCP  
= external tool/API connection
```

```
Script
= deterministic local automation
```

Recommended `code-review` skill:

```
---
name: code-review
description: Review code changes for correctness, edge cases, style, and project safety.
---

# Code Review Skill

## Steps

1. Inspect changed files.
2. Check correctness.
3. Check edge cases.
4. Check project rules from AGENTS.md.
5. Check dependency changes.
6. Check whether verification commands were run.

## Output Format

-Critical issues
-Minor issues
-Suggested fixes
-Test status
```

Recommended `safe-refactor` skill:

```
---
name: safe-refactor
description: Safely refactor code without changing behavior.
---

# Safe Refactor Skill

## Rules

-Preserve behavior.
-Do not change public function names unless required.
-Do not change file paths, output names, or data formats.
-Make small, reversible changes.
-Run tests or startup checks after each major change.
```

Required Output

- What was refactored
- Why behavior should be unchanged
- Files changed
- Verification result

Recommended `git-backup` skill:

```
---
name: git-backup
description: Prepare safe Git backup workflow for Google Drive source and GitHub backup/
deployment projects.
---

# Git Backup Skill

## Workflow

1. Check current branch.
2. Check working tree status.
3. Review changed files.
4. Do not force push.
5. Do not rewrite history.
6. Prepare a short commit message only if requested.

## Safety

- Never commit raw data unless explicitly requested.
- Never delete untracked files automatically.
- Warn if large files, raw data, or binary outputs are staged.
```

11. MCP vs Skill Distinction

The program documentation should explain the difference clearly.

```
MCP
= gives the AI access to external tools, APIs, files, databases, GitHub,
Google Drive, etc.

Skill
= tells the AI how to perform a repeatable workflow.

AGENTS.md
```

= tells the AI what rules always apply in this project.

TASK.md

= tells the AI what to do now.

Script

= deterministic local automation for repeated tasks.

Artifact

= output used to review what the AI did, such as plan, diff, walkthrough, and verification result.

Practical priority:

1. AGENTS.md
2. TASK.md
3. CONTEXT.compact.md
4. SKILL.md
5. scripts
6. MCP only when external access is necessary

12. Context Converter Modes

The program should support two conversion modes.

Mode A: Python-only rule-based conversion

Purpose:

Fast, offline, deterministic conversion without AI.

Can do:

- sentence splitting
- keyword-based classification
- duplicate removal
- filler removal
- Markdown rendering
- project tree generation
- file filtering

Cannot do well:

- semantic compression
- accurate translation
- implicit rule extraction
- context importance judgment

Mode B: Local LLM-based conversion

Purpose:

Higher-quality semantic compression and Korean-to-English agent-facing transformation.

Recommended pipeline:

```
raw Korean notes
↓
local LLM
↓
structured JSON
↓
Python validation
↓
Markdown rendering
```

Recommended local LLM use:

```
Ollama + Korean-capable local model
JSON schema output
temperature 0
short bullet output
no invented facts
```

The LLM should output JSON first, not Markdown directly.

Recommended JSON fields:

```
{
  "project": [],
  "environment": [],
  "workflow": [],
  "rules": [],
  "forbidden": [],
  "verification": [],
```

```
"output_format": []  
}
```

13. Program Feature Requirements

The final program should include these features.

Basic features

- Select project folder
- Select raw context file
- Generate AGENTS.md
- Generate TASK.md template
- Generate CONTEXT.compact.md
- Generate CONTEXT_FOR_AI.md
- Generate skill templates
- Generate project tree summary
- Exclude large or dangerous folders
- Save all outputs to selected project folder

Advanced features

- Local LLM conversion mode
- Python-only conversion mode
- token estimate
- duplicate rule detection
- forbidden-action checker
- raw data folder detector
- Google Drive sync-risk warning
- Git status summary
- large file warning
- skill template manager
- agent prompt generator

GUI requirements

- Project folder selector
- Raw context file selector
- Output folder selector
- Mode selector:
 - Rule-based
 - Local LLM
- Checkboxes:
 - Generate AGENTS.md

- Generate TASK.md
- Generate CONTEXT.compact.md
- Generate CONTEXT_FOR_AI.md
- Generate skills
- Include project tree
- Exclude raw data folders
- Start button
- Stop button
- Progress bar
- Status log
- Preview panel

14. Folder Exclusion Rules

The program should exclude these by default when building project trees or context bundles:

```
.git  
.venv  
venv  
__pycache__  
node_modules  
data  
raw  
backup  
archive  
results/raw  
dist  
build  
.cache
```

The program should warn before including:

```
large binary files  
raw experiment data  
image datasets  
video files  
model weights  
database files  
Google Drive conflict files
```

15. Token Efficiency Rules

The converter should apply these rules:

- Remove repetition.
- Remove emotional background.
- Remove unnecessary personal explanation.
- Convert long sentences into short bullets.
- Prefer key-value format.
- Preserve file paths exactly.
- Preserve commands exactly.
- Preserve forbidden actions clearly.
- Put stable rules in AGENTS.md.
- Put current task only in TASK.md.
- Put background only in CONTEXT.compact.md.
- Put repeatable procedures in SKILL.md.

Bad input style:

나는 구글 드라이브에서 개발하고 있고, 깃허브는 백업으로 쓰고,
로컬에서는 받아서 실행하는데, AI가 이걸 헛갈리면 안 되고...

Good output style:

Workflow

- Source of truth: Google Drive mirrored workspace.
- GitHub: backup, version tracking, deployment.
- Runtime machines: update through ``git pull``.
- Do not treat runtime folders as source of truth.

16. Scientific / Research Project Safety Rules

Because the user often works with MOKE, Kerr image analysis, hysteresis data, NanoMOKE3 data, and raw experimental files, the program should support scientific safety templates.

Recommended rules:

Scientific Analysis Rules

- Treat raw MOKE/Kerr/NanoMOKE files as read-only.
- Do not change H-field sign conventions without explicit approval.
- Do not change normalization formulas silently.

- Do not change fitting logic silently.
- Do not average datasets with different field sequences unless explicitly requested.
- Do not change calibration, color mapping, ROI handling, or background correction logic unless the task specifically asks for it.
- Preserve original filenames in derived output metadata.

17. Recommended User Workflow

The program should support this workflow:

1. User writes rough Korean notes in `CONTEXT.raw.ko.md`.
2. User opens the program.
3. User selects the project folder.
4. User selects conversion mode.
5. Program generates `CONTEXT.compact.md`.
6. Program generates `AGENTS.md` if missing or updates selected sections.
7. Program generates `TASK.md` template.
8. Program generates skill templates.
9. Program builds `CONTEXT_FOR_AI.md`.
10. User gives `CONTEXT_FOR_AI.md` to Codex or Antigravity.
11. AI performs the task under fixed workflow rules.
12. User reviews diff, verification result, and final report.
13. User pushes to GitHub manually when ready.
14. Runtime machines update through ``git pull``.

18. Final Design Principle

The program should not try to make the AI “smarter.”

It should make the project environment harder to misunderstand.

Core principle:

Do not give the AI more context.
Give the AI cleaner context.

Final summary:

`AGENTS.md`
= permanent law

TASK.md

= current command

CONTEXT.compact.md

= compressed background

SKILL.md

= repeatable workflow manual

MCP

= external tool connection

Script

= deterministic automation

CONTEXT_FOR_AI.md

= final input bundle for AI agents